

Ayush Katiyar

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SUMMARY

Software engineer with 2+ years at Netradyne writing Java automation (TestNG, RestAssured) against production microservices and debugging service, data and event layers end-to-end. Builds production-grade Java backend systems in parallel — a payment gateway with transactional outbox and idempotent processing, a microservices AI platform on Kubernetes, and a booking backend with pessimistic locking.

SKILLS

Languages: Java, SQL, Python, JavaScript

Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT, Microservices, Spring Cloud (Eureka, Config, Gateway), Kafka

Databases & Caching: PostgreSQL, MySQL, Redis

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Kubernetes, GitHub Actions (CI/CD), Maven, Git

Fundamentals: Data Structures & Algorithms, OOP, System Design, Design Patterns, Concurrency Control

Testing: JUnit, Mockito, TestNG, RestAssured, REST API Automation, Regression Automation

Tools: IntelliJ IDEA, Postman, Jira

PROJECTS

Payment Gateway & Ledger Service

Tech: Java, Spring Boot, PostgreSQL, Kafka, Redis, Docker

- Designed a payment lifecycle state machine (Created – Captured – Settled/Refunded) with a full transition audit trail.
- Implemented the Transactional Outbox pattern for reliable Kafka event delivery across services, preventing dual-write inconsistencies.
- Built a Redis-backed idempotency filter with request locking and response replay to block duplicate charges.
- Built a PCI-inspired card vault with AES-GCM envelope encryption, tokenization, and a Redis Lua rate limiter.

Distributed AI App-Builder Platform

Tech: Java, Spring Boot, Spring Cloud, Kafka, Redis, Kubernetes, MinIO, OpenAI

- Designed a microservices platform using Spring Boot & Spring Cloud (Eureka, Config, Gateway).
- Integrated Spring AI + OpenAI with tool-calling and RAG-style context injection for code generation.
- Used Kafka with a Saga pattern & idempotent consumers for cross-service consistency.
- Provisioned live app previews via dynamic Kubernetes pod allocation, MinIO sync, and Redis caching.

Hotel Booking & Dynamic Pricing Backend

Tech: Java, Spring Boot, Spring Security (JWT), PostgreSQL, Stripe

- Built a booking backend with Spring Boot, Spring Security (JWT), PostgreSQL, and Stripe payments.
- Designed a dynamic pricing engine (Decorator pattern) refreshed hourly via a `@Scheduled` job.
- Solved double-booking race conditions using JPA pessimistic locking with atomic bulk updates.

EXPERIENCE

Software Engineer

Netradyne Technology India Private Limited, Bengaluru, India | Jan 2025 – Present

- Built and maintained Java API automation with TestNG and RestAssured against the Speeding V2 microservices, working alongside backend engineers on root-cause analysis across service and data layers; surfaced 250+ defects pre-release.
- Debugged the Alert Geofence backend module end-to-end — REST endpoints, database queries and event flows — isolating 100+ defects and their underlying causes in service code.
- Designed and built end-to-end API regression suites in TestNG, cutting manual test effort by 60% and shortening release cycles.

Associate Software Engineer

Netradyne Technology India Private Limited, Bengaluru, India | Jul 2024 – Dec 2024

- Automated validation of service-layer logic and data integrity across backend REST APIs in multiple environments using RestAssured.
- Investigated API failures and data inconsistencies through backend logs, service responses, and database-level analysis.
- Promoted to Software Engineer within 6 months for high-impact contributions to release quality.

EDUCATION

BMS College of Engineering | B.E. in Information Science & Engineering | 2020 – 2024 | CGPA: 8.73

ACHIEVEMENTS

- Built and deployed 3 production-style backend systems using Spring Boot, Kafka, Redis and PostgreSQL.
- Solved 500+ data structures and algorithms problems — 200+ on LeetCode and 300+ on GeeksforGeeks.
- Promoted to Software Engineer within 6 months at Netradyne.